



TATA STEEL · JAMSHEDPUR · AGENTIC AI PLATFORM

Industrial Maintenance

Transforming Equipment Data, Maintenance Knowledge,
Inventory Intelligence, and Engineering Decisions
into One Unified System.

An integrated industrial maintenance platform combining AI-assisted
troubleshooting, maintenance knowledge retrieval, inventory intelligence, workflow
automation, safety verification, equipment monitoring, and engineering decision
support.

Access Platform

View Architecture

AI
FEATURES

AGENT
LAYERS

LOCAL ·
PRIVATE

CLOUD
DEPENDENCY

CONTINUOUS
LEARNING

Multi-Agent Industrial Intelligence Architecture

A two-layer AI architecture that separates rapid operational assistance from deep industrial reasoning while maintaining complete context transfer and continuous learning.



Sandbox Agent

Small Language Model (SLM)

Fast operational workspace for engineers.

- Engineering Discussions
- Requirement Gathering
- Troubleshooting Support
- Maintenance Planning
- Inventory Questions
- Workflow Design
- Context Collection
- Agent Escalation



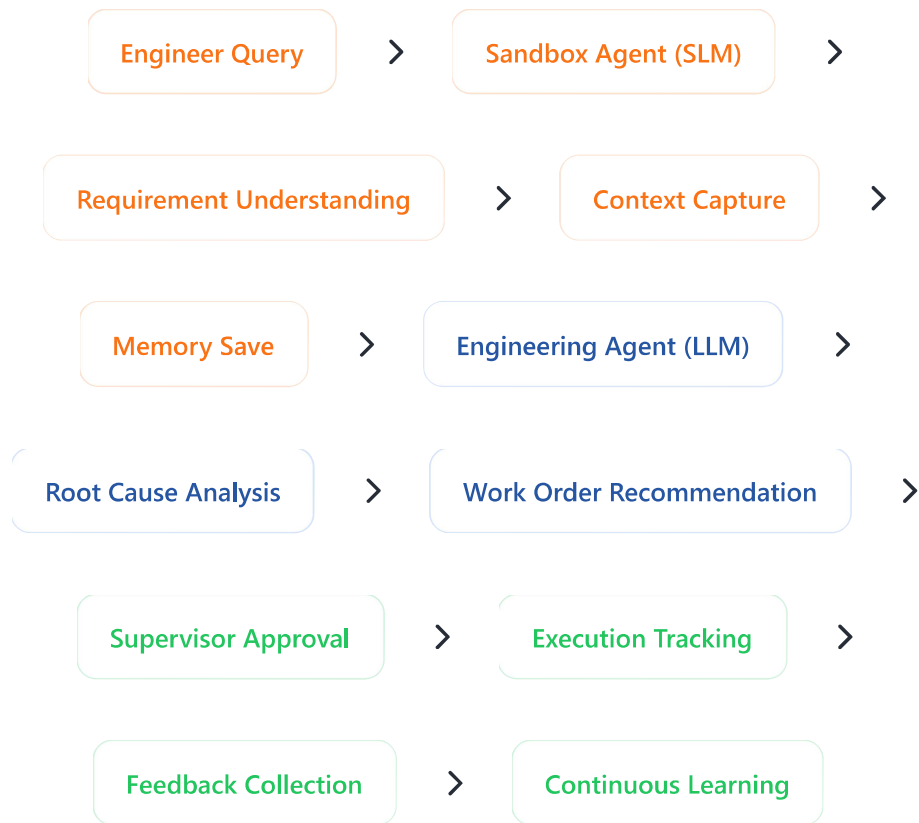
Engineering Agent

Large Language Model (LLM)

Advanced industrial reasoning and execution workspace.

- Root Cause Analysis
- Maintenance Diagnostics
- Equipment Health Evaluation
- Work Order Recommendation
- Safety Verification
- Inventory Intelligence
- Maintenance Knowledge Search
- Escalation Recommendations
- Maintenance History Analysis
- Engineering Decision Support

Multi-Agent Transfer Flow



ARCHITECTURE

Industrial AI Intelligence Layer

The platform combines multiple intelligence layers into a single industrial maintenance ecosystem.

SLM Layer

Fast operational assistance

LLM Layer

Advanced engineering reasoning

Knowledge Layer

Maintenance manuals

SOPs

Procedures

Workflow Layer

Work Orders

Approvals

Escalations

Data Layer

Users & Equipment

Inventory

Maintenance History & Audit Logs

Learning Layer

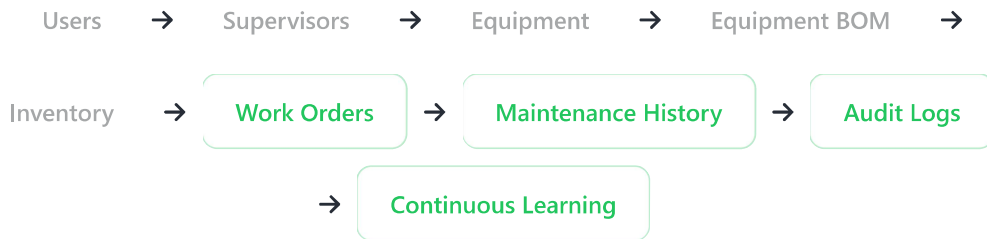
Feedback

Outcomes

Continuous Improvement

Connected Industrial Knowledge Network

Every component of the platform is connected through a unified industrial database architecture enabling intelligent querying, traceability, maintenance analytics, and operational decision support.



PLATFORM CAPABILITIES

Everything in One Platform

No switching between tools. Every industrial intelligence capability connected.



AI Engineering Assistant



Maintenance Knowledge Search



Root Cause Analysis



Equipment Health Monitoring



Inventory Intelligence



Work Order Automation



Supervisor Escalation Workflow



Safety Verification



Maintenance History Intelligence



Audit & Compliance Tracking



Industrial Decision Support



Continuous Learning & Feedback

Continuous Learning & Feedback Engine

The platform improves through operational feedback and maintenance outcomes.

LEARNING SOURCES

Engineer Feedback

Supervisor Reviews

Maintenance Results

Equipment Performance

Inventory Usage

Escalation Outcomes

Work Order Completion Data

BENEFITS

- ✓ Improved Recommendations
- ✓ Better Diagnostics
- ✓ Reduced Downtime
- ✓ Smarter Decision Support
- ✓ Knowledge Retention

ACCESS PORTAL

Secure Login

Enter your credentials to access the platform

Access Platform

Select a role to enter the Industrial Maintenance
Intelligence Platform



Manager Portal

Employee ID: MGR001

Password: Manager@123

➡ Login as Manager



Supervisor Portal

Employee ID: SUP001

Password: Supervisor@123

➡ Login as Supervisor

Engineer Portal

Employee ID: ENG001

Password: Engineer@123

 Login as Engineer

Or Login with your own credentials

ENG011

.....

 Login

Maintenance Wizard

Industrial Maintenance Intelligence Platform

Developed By: Vikhil Ravella

 Powered By: Ollama Local Models

Role-Based Access Control

Audit Logging Enabled

Industrial Operations Platform